MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

COMPUTER AND COMMUNICATION TECHNOLOGY

(ES-WRMD 1106/ICT 1103)

END OF SEMESTER EXAMINATION

TIME ALLOWED: 2 HRS **DATE : 7th SEPTEMBER , 2017**

INSTRUCTIONS

i. This paper contains **two** pages. Please check.
ii. There are **five questions** in this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

organisational

Question One

1. List **four** distinct parts of Information Technology systems in an organization. **(4 Marks)**
2. Mention and explain **four** impacts of Information Technology in an organization. **(8 Marks)**
3. What is the difference between Information Systems and Information Technology? **(4 Marks)**
4. Discuss **four** threats of computer and communication systems. **(8 Marks)**

Question Two

1. Mention **two** principle benefits of computer networks. **(2 Marks)**
2. Mention and discuss **three** classifications of computer networks. **(9 Marks)**
3. Define the term Network Topology. **(2 Marks)**
4. What is the difference between Logical and Physical topologies? **(2 Marks)**
5. Explain any **three** types of topologies in computer networks. **(9 Marks)**

Question Three

Most organisations utilise a variety of information and usually have more than one information system and below is the list of those information systems.

1. Give **three** examples of each of the following:
   a. Executive Support Systems/Executive Information Systems **(3 Marks)**
   b. Management Information Systems **(3 Marks)**
   c. Knowledge Management Systems **(3 Marks)**
   d. Transaction Processing Systems **(3 Marks)**
   e. Office Automation Systems **(3 Marks)**
2. State **four** purposes of doing analysis and requirements capture in Systems Development Life Cycle. **(4 Marks)**

Question Four

1). Explain the following terms as used in data communication in terms of direction of Transmission Flow (Data Flow) and give **one example** for each. **(15 marks)**

a. Simplex Mode
b. Half Duplex
c. Full Duplex
d. Asynchronous
e. Synchronous.

2). List any **four** components of data communications system. **(4 marks)**

Question Five

1. What are Numbering systems? **(2 Marks)**
2. What is the difference between Encoding and Decoding? **(2 Marks)**
3. Convert the following:
   i. 1010001011011110_2 to Hexadecimal number systems **(2 Marks)** $A2DE_{16}$
   ii. FF_{16} to binary number. **(2 Marks)**
   iii. $E5_{16}$ to octal number system. **(2 Marks)** 347
   iv. 231_{10} to octal number system. **(2 Marks)**
   v. 121336_8 to hexadecimal number systems. **(2 Marks)** A2DE

(2) 4

$(1 \times 16)_{16}$

64	32	16	8	4	2	1
			1	1	0	1
			1	1	0	0
			1	0	1	0
			0	1	0	0

A2DE

64	32	16	8	4	2	1

64	32	16	8	4	2	1
			1	0	1	1
			0	0	1	0
			0	0	0	0
			0	0	0	0

END OF PAPER